

Comprehensive Multi-Dataset, Multi-Model Comparison

Metric	LNSSM (GEFCom2012)	LNSSM (GEFCom2014)	QRF [5]	Persistence	Industry Target
Pinball Loss (99Q)	0.0806	0.2069	0.1003	0.1192	< 0.10
CRPS	0.169	—	0.211	—	< 0.15
RMSE (p.u.)	0.280	0.312	0.264	0.294	< 0.20
MAE (p.u.)	0.209	0.238	0.195	0.238	< 0.15
R ² (+1h)	+0.60	+0.46	+0.42	< 0	> 0.65
PICP (80% CI)	45.5%	69.9%	0%[(+)]	—	70–90%
PINAW (80% CI)	0.299	0.342	—	—	< 0.25
Reliability Dev.	23.8%	12.4%	80%[(+)]	—	< 10%
Parameters	412K	396K	99x50T	0	—
Training Time	2.1 h	0.3 h	5.2 h	—	< 4 h
Inference (per window)	5 ms	4 ms	120 ms	0 ms	< 10 ms

[(+)]QRF PI metrics degraded by quantile crossing. All power metrics on [0,1]. GEFCom2012 = 7-farm cross-train. GEFCom2014 = Zone1 single-site.